

# Kai Zhang

**Email:** kai.zhang@neocognition.io

**Personal Website:** <https://drogozhang.github.io>

I am interested in improving foundation models and agents with data solutions. I have published papers in top NLP/CV/ML venues, many of which have been nominated for or won Best Paper Awards (3), or have been featured as Oral (8) and Spotlight (3) presentations. My work has 6,000 citations and an h-index of 26.

## EDUCATION

### The Ohio State University

Ph.D. in Computer Science and Engineering

**Advisor:** Yu Su and Huan Sun

Columbus, OH, USA

Aug 2022-May 2026 (ETA)

### Zhejiang University City College

B.E. in Computer Science and Technology

Zhejiang, China

Sept 2016-Jun 2020

## WORK EXPERIENCE

### NeoCognition Inc.

Founding Member of Technical Staff

Palo Alto, CA, USA

Sept 2025-Present

### Meta SuperIntelligence Labs

Research Scientist Intern; Collaborators: Jiachen Zhu, Shuyan Zhou and Jason Weston

New York, NY, USA

May 2025-Aug 2025

### Microsoft Research

Research Intern; Host: Chenglong Wang and Jianwei Yang

Redmond, WA, USA

May 2024-Aug 2024

### Google DeepMind

Student Researcher; Host: Yi Luan, Hexiang (Frank) Hu, and Ming-Wei Chang

Kirkland, WA, USA

Aug 2023-Mar 2024

### Microsoft Research

Research Intern; Host: Chongyang Tao and Tao Shen

Beijing, China

Mar 2022-July 2022

### Minds, Machines, and Society Group, Dartmouth College

Research Intern; Advisor: Soroush Vosoughi

Hanover, NH, USA

Jun 2020-Sept 2020

### THU NLP Group, Tsinghua University

Research Assistant; Advisor: Zhiyuan Liu

Beijing, China

Nov 2019-Mar 2021

## SELECTED PUBLICATIONS

(† indicates my mentee. See Google Scholar)

### 1. Is Your LLM Secretly a World Model of the Internet? Model-Based Planning for Web Agents

Yu Gu\*, Kai Zhang\*, Yuting Ning\*, Boyuan Zheng\*, Boyu Gou, Tianci Xue, Cheng Chang, Sanjari Srivastava, Yanan Xie, Peng Qi, Huan Sun, and Yu Su

In TMLR, 2025 (\*: Equal Contribution)

### 2. MagicLens: Self-Supervised Image Retrieval with Open-Ended Instructions

Kai Zhang, Yi Luan, Hexiang Hu, Kenton Lee, Siyuan Qiao, Wenhui Chen, Yu Su, and Ming-Wei Chang

In ICML (**Oral**), 2024.

**Data Used in Gemini Embedding; Model Used in Snap, Pinterest, and Google**

### 3. TravelPlanner: A Benchmark for Real-World Planning with Language Agents

Jian Xie†\*, Kai Zhang\*, Jiangjie Chen, Tinghui Zhu, Renze Lou, Yuandong Tian, Yanghua Xiao, and Yu Su

In ICML (**Spotlight**), 2024. (\*: Equal Contribution)

### 4. MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI

Xiang Yue\*, Yuansheng Ni\*, Kai Zhang\*, Tianyu Zheng\*, Ruoyi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, Ruibin Yuan, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, and Wenhui Chen

In CVPR (**Oral; Best Paper Finalist**), 2024. (\*: Core Contribution)

**2M Total Downloads; Used by Google DeepMind, OpenAI, Anthropic, xAI, NVIDIA, Meta, and Microsoft**

5. **Adaptive Chameleon or Stubborn Sloth: Revealing the Behavior of Large Language Models in Knowledge Conflicts**

Jian Xie†\*, Kai Zhang\*, Jiangjie Chen, Renze Lou, and Yu Su

In ICLR (**Spotlight**), 2024. (\*: Equal Contribution)

6. **MagicBrush: A Manually Annotated Dataset for Instruction-Guided Image Editing**

Kai Zhang\*, Lingbo Mo\*, Wenhui Chen, Huan Sun, and Yu Su

In NeurIPS Datasets and Benchmarks, 2023. (\*: Equal Contribution)

**300K Total Downloads; Data Used in Imagen (Google), Emu Edit (Meta), and Various Open Models**

7. **Aligning Instruction Tasks Unlocks Large Language Models as Zero-Shot Relation Extractors**

Kai Zhang, Bernal Jiménez Gutiérrez, and Yu Su

In Findings of ACL, 2023.

## OTHER PUBLICATIONS

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1. **ARM: Adaptive Reasoning Model**

Siye Wu, Jian Xie, Yikai Zhang, Aili Chen, Kai Zhang, Yu Su, and Yanghua Xiao

In NeurIPS (**Spotlight**), 2025.

2. **Mind2Web 2: Evaluating Agentic Search with Agent-as-a-Judge**

Boyu Gou, Zanming Huang, Yuting Ning, Yu Gu, Michael Lin, Weijian Qi, Andrei Kopanov, Botao Yu, Bernal Jiménez Gutiérrez, Yiheng Shu, Chan Hee Song, Jiaman Wu, Shijie Chen, Hanane Nour Moussa, Tianshu Zhang, Jian Xie, Yifei Li, Tianci Xue, Zeyi Liao, Kai Zhang, Boyuan Zheng, Zhaowei Cai, Viktor Rozgic, Morteza Ziyadi, Huan Sun, and Yu Su

In NeurIPS Datasets and Benchmark, 2025.

3. **CPathAgent: An Agent-based Foundation Model for Interpretable High-Resolution Pathology Image Analysis Mimicking Pathologists' Diagnostic Logic**

Yuxuan Sun, Yixuan Si, Chenglu Zhu, Kai Zhang, Zhongyi Shui, Bowen Ding, Tao Lin, and Lin Yang

In NeurIPS, 2025.

4. **PathBench: Advancing the Benchmark of Large Multimodal Models for Pathology Image Understanding at Patch and Whole Slide Level**

Yuxuan Sun, Hao Wu, Chenglu Zhu, Yixuan Si, Qizi Chen, Yunlong Zhang, Kai Zhang, Jingxiong Li, Jiatong Cai, Yuhang Wang, Lin Sun, Tao Lin, and Lin Yang

In IEEE Transactions on Medical Imaging, 2025.

5. **MMMU-Pro: A More Robust Multi-discipline Multimodal Understanding Benchmark**

Xiang Yue, Tianyu Zheng, Yuansheng Ni, Yubo Wang, Kai Zhang, Shengbang Tong, Yuxuan Sun, Botao Yu, Ge Zhang, Huan Sun, Yu Su, Wenhui Chen, and Graham Neubig

In ACL, 2025.

6. **CPath-Omni: A Unified Multimodal Foundation Model for Patch and Whole Slide Image Analysis in Computational Pathology**

Yuxuan Sun, Yixuan Si, Chenglu Zhu, Xuan Gong, Kai Zhang, Pingyi Chen, Ye Zhang, Zhongyi Shui, Tao Lin, and Lin Yang

In CVPR, 2025.

7. **AAAR-1.0: Assessing AI's Potential to Assist Research**

Renze Lou, Hanzi Xu, Sijia Wang, Jiangshu Du, Ryo Kamoi, Xiaoxin Lu, Jian Xie, Yuxuan Sun, Yusen Zhang, Jihyun Janice Ahn, Hongchao Fang, Zhuoyang Zou, Wenchao Ma, Xi Li, Kai Zhang, Congying Xia, Lifu Huang, and Wenpeng Yin

In ICML, 2025; AI4Research@AAAI (**Best Paper Award**), 2025.

8. **PathGen-1.6M: 1.6 Million Pathology Image-Text Pairs Generation through Multi-Agent Collaboration**

Yuxuan Sun, Yunlong Zhang, Yixuan Si, Chenglu Zhu, Zhongyi Shui, Kai Zhang, Jingxiong Li, Xingheng Lyu,

Tao Lin, and Lin Yang

In ICLR (**Oral**), 2025.

9. **MuirBench: A Comprehensive Benchmark for Robust Multi-image Understanding**  
Fei Wang, Xingyu Fu, James Y Huang, Zekun Li, Qin Liu, Xiaogeng Liu, Mingyu Derek Ma, Nan Xu, Wenxuan Zhou, Kai Zhang, Tianyi Lorena Yan, Wenjie Jacky Mo, Hsiang-Hui Liu, Pan Lu, Chunyuan Li, Chaowei Xiao, Kai-Wei Chang, Dan Roth, Sheng Zhang, Hoifung Poon, and Muhao Chen  
In ICLR, 2025.
10. **Revealing the Barriers of Language Agents in Planning**  
Jian Xie, Kexun Zhang, Jiangjie Chen, Siyu Yuan, Kai Zhang, Yikai Zhang, Lei Li, and Yanghua Xiao  
In NAACL, 2025.
11. **PathMMU: A Massive Multimodal Expert-Level Benchmark for Understanding and Reasoning in Pathology**  
Yuxuan Sun, Hao Wu, Chenglu Zhu, Sunyi Zheng, Qizi Chen, Kai Zhang, Yunlong Zhang, Xiaoxiao Lan, Mengyue Zheng, Jingxiong Li, Xinheng Lyu, Tao Lin, and Lin Yang  
In ECCV (**Oral; Best Paper Finalist**), 2024.
12. **Deductive Beam Search: Decoding Deducible Rationale for Chain-of-Thought Reasoning**  
Tinghui Zhu†\*, Kai Zhang\*, Jian Xie, and Yu Su  
In COLM, 2024. (\*: Equal Contribution)
13. **How Easily do Irrelevant Inputs Skew the Responses of Large Language Models?**  
Siye Wu, Jian Xie, Jiangjie Chen, Tinghui Zhu, Kai Zhang, and Yanghua Xiao  
In COLM, 2024.
14. **MUFFIN: Curating Multi-Faceted Instructions for Improving Instruction-Following**  
Renze Lou†, Kai Zhang, Jian Xie, Yuxuan Sun, Janice Ahn, Hanzi Xu, Yu Su, and Wenpeng Yin  
In ICLR, 2024.
15. **ImagenHub: Standardizing the Evaluation of Conditional Image Generation Models**  
Max Ku, Tianle Li, Kai Zhang, Yujie Lu, Xingyu Fu, Wenwen Zhuang, and Wenhui Chen  
In ICLR, 2024.
16. **Large Language Model Instruction Following: A Survey of Progresses and Challenges**  
Renze Lou†, Kai Zhang, and Wenpeng Yin  
In Computational Linguistics, 2024.
17. **CIF-Bench: A Chinese Instruction-Following Benchmark for Evaluating the Generalizability of Large Language Models**  
Yizhi Li, Ge Zhang, Xingwei Qu, Jiali Li, Zhaoqun Li, Zekun Wang, Haoyuan Li, Ruibin Yuan, Yi Ma, Kai Zhang, Wangchunshu Zhou, Yiming Liang, Lei Zhang, Lei Ma, Jiajun Zhang, Zuowen Li, Stephen W. Huang, Chenghua Lin, Wenhui Chen, and Jie Fu  
In Findings of ACL, 2024.
18. **UMIE: Unified Multimodal Information Extraction with Instruction Tuning**  
Lin Sun\*, Kai Zhang\*, Qingyuan Li†\*, and Renze Lou  
In AAI (**Oral**), 2024. (\*: Equal Contribution)
19. **PathAsst: Redefining Pathology through Generative Foundation AI Assistant for Pathology**  
Yuxuan Sun, Chenglu Zhu, Sunyi Zheng, Kai Zhang, Zhongyi Shui, Xiaoxuan Yu, Yizhi Zhao, Honglin Li, Yunlong Zhang, Ruojia Zhao, Xinheng Lyu, and Lin Yang  
In AAI (**Oral**), 2024.
20. **Automatic Evaluation of Attribution by Large Language Models**  
Xiang Yue, Boshi Wang, Ziru Chen, Kai Zhang, Yu Su, and Huan Sun  
In Findings of EMNLP, 2023.
21. **UnifieR: A Unified Retriever for Large-Scale Retrieval**  
Tao Shen, Xiubo Geng, Chongyang Tao, Can Xu, Guodong Long, Kai Zhang, and Daxin Jiang  
In KDD, 2023.

22. **LED: Lexicon-Enlightened Dense Retriever for Large-Scale Retrieval**  
 Kai Zhang, Chongyang Tao, Tao Shen, Can Xu, Xiubo Geng, Binxing Jiao, and Daxin Jiang  
 In WWW (**Oral**), 2023.
23. **ChatDoctor: A Medical Chat Model Fine-Tuned on a Large Language Model Meta-AI (LLaMA) Using Medical Domain Knowledge**  
 Yunxiang Li, Zihan Li, Kai Zhang, Ruilong Dan, and You Zhang  
 In Cureus, 2023.
24. **Contributions of Transformer Attention Heads in Multi- and Cross-lingual Tasks**  
 Weicheng Ma\*, Kai Zhang\*, Renze Lou, Lili Wang, and Soroush Vosoughi  
 In ACL (**Oral**), 2021. (\*: Equal Contribution)
25. **GradTS: A Gradient-Based Automatic Auxiliary Task Selection Method Based on Transformer Network**  
 Weicheng Ma, Renze Lou, Kai Zhang, Lili Wang, and Soroush Vosoughi  
 In EMNLP, 2021.
26. **Open Hierarchical Relation Extraction**  
 Kai Zhang\*, Yuan Yao\*, Ruobing Xie, Xu Han, Zhiyuan Liu, Fen Lin, Leyu Lin, and Maosong Sun  
 In NAACL-HLT, 2021. (\*: Equal Contribution)
27. **Unified and Incremental SimRank: Index-free Approximation with Scheduled Principle**  
 Fanwei Zhu, Yuan Fang, Kai Zhang\*, Kevin Chen-Chuan Chang, Hongtai Cao, Zhen Jiang, and Minghui Wu  
 In IEEE TKDE, 2021. (\*: First Student Author)
28. **Adversarial Language Games for Advanced Natural Language Intelligence**  
 Yuan Yao, Haoxi Zhong, Zhengyan Zhang, Xu Han, Xiaozhi Wang, Kai Zhang, Chaojun Xiao, Guoyang Zeng, Zhiyuan Liu, and Maosong Sun  
 In AAI, 2021.
29. **RpBERT: A Text-image Relation Propagation-based BERT Model for Multimodal NER**  
 Lin Sun\*, Jiquan Wang†\*, Kai Zhang, Fangsheng Weng, and Yindu Su  
 In AAI, 2021. (\*: Equal Contribution)
30. **TOI-CNN: A Solution of Information Extraction on Chinese Insurance Policy**  
 Lin Sun\*, Kai Zhang\*, Fule Ji, and Zhenhua Yang  
 In NAACL-HLT, 2019. (\*: Equal Contribution with Advisor)

**SELECTED HONORS & AWARDS**

|                                                               |      |
|---------------------------------------------------------------|------|
| Best Paper Award, AI4Research Workshop@AAAI                   | 2025 |
| Best Paper Finalist, ECCV                                     | 2024 |
| Best Paper Finalist, CVPR                                     | 2024 |
| Valedictorian (Top 1), Zhejiang University City College       | 2020 |
| National Scholarship (Top 1%), Ministry of Education of China | 2019 |

**INVITED TALKS**

|                                                          |
|----------------------------------------------------------|
| <b>Agent Learning via Early Experience</b>               |
| ByteDance Seed                                           |
| Intuit Research                                          |
| NLP Academic Exchange Platform                           |
| <b>MagicLens: Next-Generation Image Retrieval Models</b> |
| AI TIME Seminar                                          |
| NLP Group, Zhejiang University                           |
| Beijing Academy of Artificial Intelligence               |
| <b>Language Agents: Grounding and Planning</b>           |
| AISOC Lab, Carnegie Mellon University                    |

## **TravelPlanner: A Benchmark for Real-World Planning with Language Agents**

Real World Journeys Group, Google

## **Aligning LLM Instruction Tasks for Zero-Shot Relation Extraction**

NLP Seminar, University of Notre Dame

AI TIME Seminar

## **MENTORING**

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Renze Lou (ZUCC B.S. → PSU Ph.D.; 2019–2022): Instruction Tuning

Jian Xie (ZUCC B.S. → FDU M.S. → OSU Ph.D.; 2019–2024): Language Agents

Yuxuan Sun (ZUCC B.S. → ZJU Ph.D.; 2019–2024): Multimodality, Medical AI

Tinghui Zhu (FDU M.S. → UCD Ph.D.; 2023–2025): Reasoning, Multimodality

## **SERVICES**

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### **Area Chair:**

**ML:** ICLR'26

### **Workshop (Co-)Organizer:**

Trust Before Use: Building Foundation Models that You Can Trust@ICCV 2025

Multi-Turn Interactions in Large Language Models@NeurIPS 2025

### **PC Member/Reviewer:**

**NLP:** ARR (since 2022); CCKS; NLPCC; EMNLP; ACL; NAACL

**CV:** CVPR; ICCV; TPAMI

**ML:** NeurIPS; ICLR; ICML; COLM

**Other:** WWW; KDD; CIKM; SDM; UIST; AAAI; TMM